

Hypothesis Testing



Class Objectives

- Developing Null and Alternative Hypotheses
- Type I and Type II Errors- Explanation
- Population Mean: Sigma Known
- Population Mean: Sigma Unknown
- Population Proportion



Hypothesis Testing

- Hypothesis testing can be used to determine whether a statement about the value of a population parameter should or should not be rejected.
- The null hypothesis, denoted by H_0 , is a tentative assumption about a population parameter
- The alternative hypothesis, denoted by H_a , is the opposite of what is stated in the null hypothesis
- The hypothesis testing procedure uses data from a sample to test the two competing statements indicated by H_0 and H_a .



Developing Null and Alternative Hypotheses

- It is not always obvious how the null and alternative hypotheses should be formulated
- Care must be taken to structure the hypotheses appropriately so that the test conclusion provides the information the researcher wants
- The context of the situation is very important in determining how the hypotheses should be stated
- In some cases it is easier to identify the alternative hypothesis first. In other cases the null is easier
- Correct hypothesis formulation will take practice



Developing Null and Alternative Hypotheses

Alternative Hypothesis as a Research Hypothesis

- Many applications of hypothesis testing involve an attempt to gather evidence in support of a research hypothesis
- In such cases, it is often best to begin with the alternative hypothesis and make it the conclusion that the researcher hopes to support
- The conclusion that the research hypothesis is true is made if the sample data provide sufficient evidence to show that the null hypothesis can be rejected



Developing Null and Alternative Hypotheses

Alternative Hypothesis as a Research Hypothesis

- Example: A new manufacturing method is believed to be better than the current method.
- Alternative Hypothesis:
 - The new manufacturing method is better.
- Null Hypothesis:
 - The new method is no better than the old method.



Developing Null and Alternative Hypotheses

- Alternative Hypothesis as a Research Hypothesis
- Example: A new bonus plan, that is developed in an attempt to increase sales
- Alternative Hypothesis:
 - The new bonus plan increase sales
- Null Hypothesis:
 - The new bonus plan does not increase sales



Developing Null and Alternative Hypotheses

- Alternative Hypothesis as a Research Hypothesis
- Example:
 - A new drug is developed with the goal of lowering Cholesterol-level more than the existing drug
- Alternative Hypothesis:
 - The new drug lowers Cholesterol-level more than the existing drug
- Null Hypothesis:
 - The new drug does not lower Cholesterol-level more than the existing drug

Developing Null and Alternative Hypotheses

- Null Hypothesis as an assumption to be challenged
- We might begin with a belief or assumption that a statement about the value of a population parameter is true
- We then using a hypothesis test to challenge the assumption and determine if there is statistical evidence to conclude that the assumption is incorrect
- In these situations, it is helpful to develop the null hypothesis first



Developing Null and Alternative Hypotheses

- Null Hypothesis as an Assumption to be Challenged
- Example:
 - The label on a milk bottle states that it contains 1000 ml
- Null Hypothesis:
 - The label is correct. $\mu \geq 1000$ ml
- Alternative Hypothesis:
 - The label is incorrect. $\mu < 1000$ ml



Null and Alternative Hypotheses about a Population Mean μ

- The equality part of the hypotheses always appears in the null hypothesis
- In general, a hypothesis test about the value of a population mean μ must take one of the following three forms (where μ_0 is the hypothesized value of the population mean)

$$H_0: \mu \geq \mu_0 \quad H_0: \mu \leq \mu_0 \quad H_0: \mu = \mu_0$$

$$H_a: \mu < \mu_0 \quad H_a: \mu > \mu_0 \quad H_a: \mu \neq \mu_0$$

One-tailed
(lower-tail)

One-tailed
(upper-tail)

Two-tailed

Null and Alternative Hypotheses

- A major hospital in Chennai provides one of the most comprehensive emergency medical services in the world
- Operating in a multiple hospital system with approximately 10 mobile medical units, the service goal is to respond to medical emergencies with a mean time of 8 minutes or less
- The director of medical services wants to formulate a hypothesis test that could use a sample of emergency response times to determine whether or not the **service goal of 8 minutes or less is being achieved.**



Null and Alternative Hypotheses

$$H_0: \mu \leq 8$$

The emergency service is meeting the response goal; no follow-up action is necessary.

$$H_a: \mu > 8$$

The emergency service is not meeting the response goal; appropriate follow-up action is necessary.

where: μ = mean response time for the population
of medical emergency requests

Type I Error

- Because hypothesis tests are based on sample data, we must allow for the possibility of errors
- A Type I error is rejecting H_0 when it is true
- The probability of making a Type I error when the null hypothesis is called the level of significance
- Applications of hypothesis testing that only control the Type I error are often called significance tests



Type II Error

- A Type II error is accepting H_0 when it is false.
- It is difficult to control for the probability of making a Type II error.
- Statisticians avoid the risk of making a Type II error by using “do not reject H_0 ” and not “accept H_0 ”.



Type I and Type II Errors

	Population Condition	
	H0 True ($\mu \leq 8$)	H0 False ($\mu > 8$)
Conclusion		
Accept H0 (Conclude $\mu \leq 8$)	Correct Decision	Type II Error
Reject H0 (Conclude $\mu > 8$)	Type I Error	Correct Decision

Three Approaches for Hypothesis Testing

- P- Value (Probability that null hypothesis is true)
- Critical Value
- Confidence Interval Value



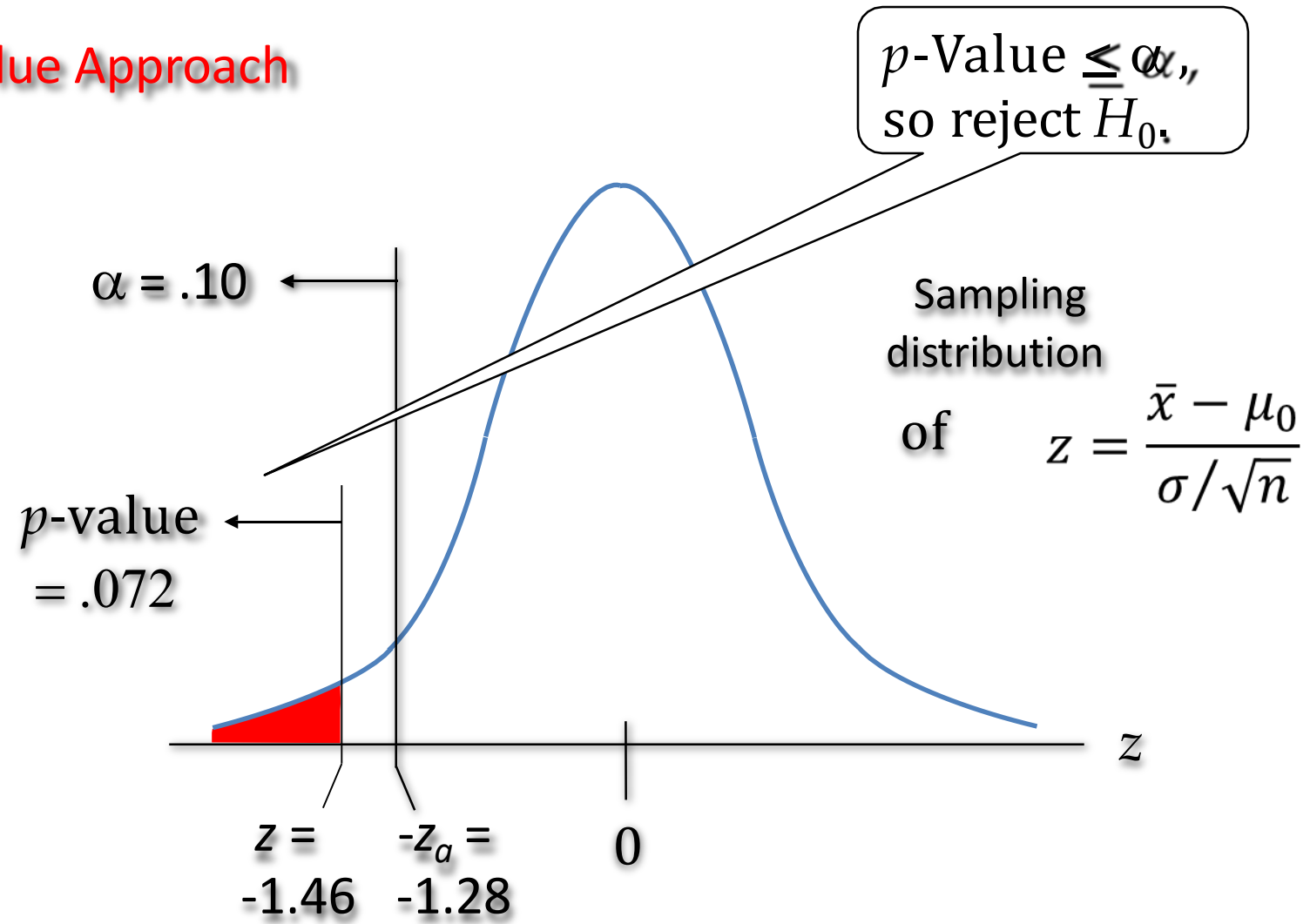
p-Value Approach to One-Tailed Hypothesis Testing

- The p -value is the probability, computed using the test statistic, that measures the support (or lack of support) provided by the sample for the null hypothesis
- If the p -value is less than or equal to the level of significance α , the value of the test statistic is in the rejection region
- Reject H_0 if the p -value $\leq \alpha$



Lower-Tailed Test About a Population Mean: σ Known

p-Value Approach



p-Value Approach

Finding P Value

```
In [3]: stats.norm.cdf(-1.46)
```

```
Out[3]: 0.07214503696589378
```

Finding p value from z value

Finding Z Value

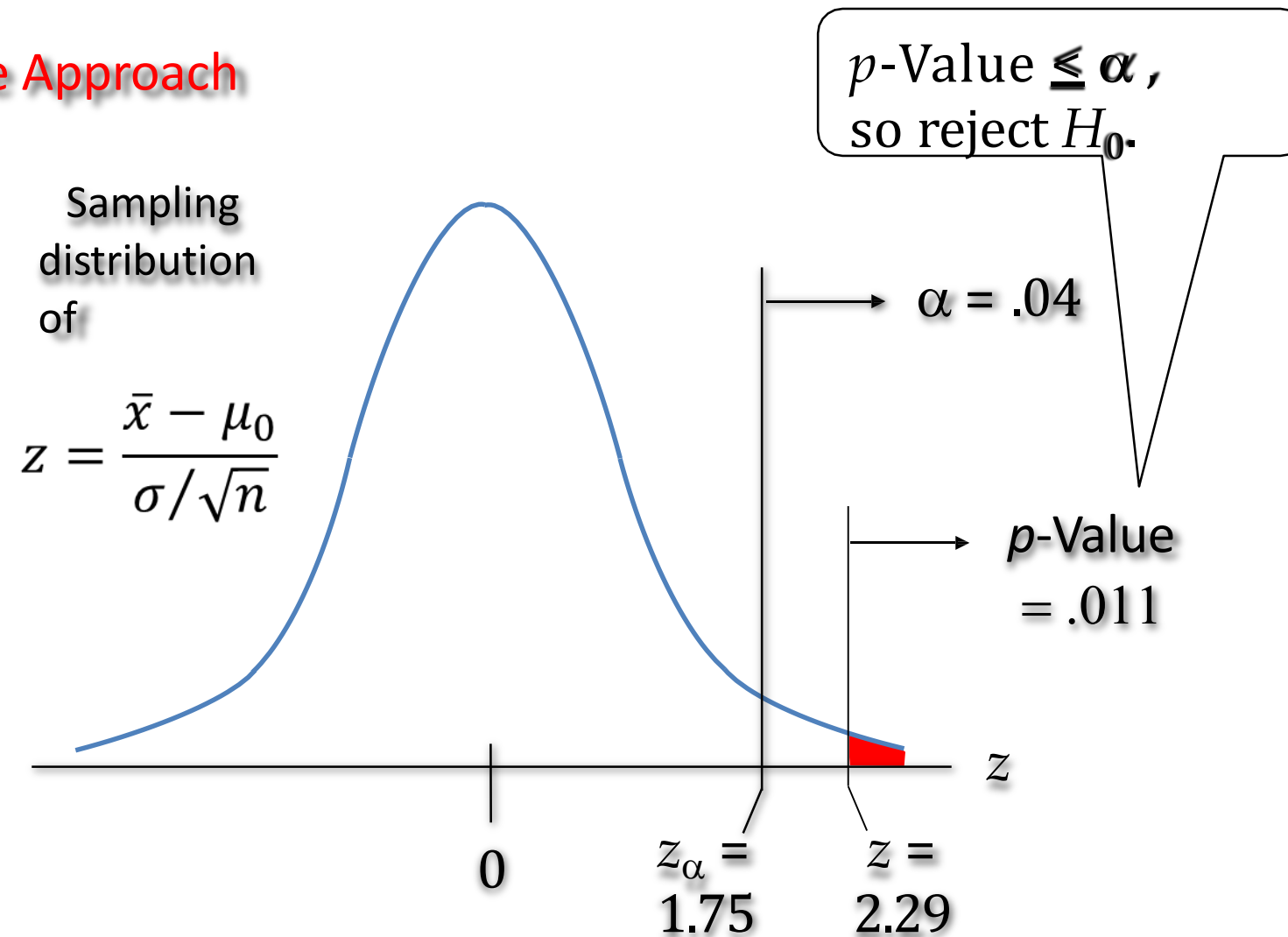
```
In [5]: stats.norm.ppf(0.1)
```

```
Out[5]: -1.2815515655446004
```

Finding z value from alpha(α) value

Upper-Tailed Test About a Population Mean : σ Known

p -Value Approach



p -Value Approach

```
In [4]: 1-stats.norm.cdf(1.75)
```

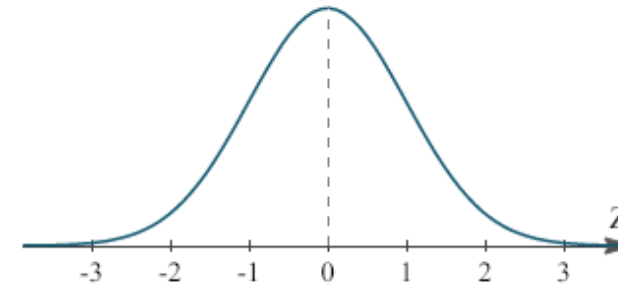
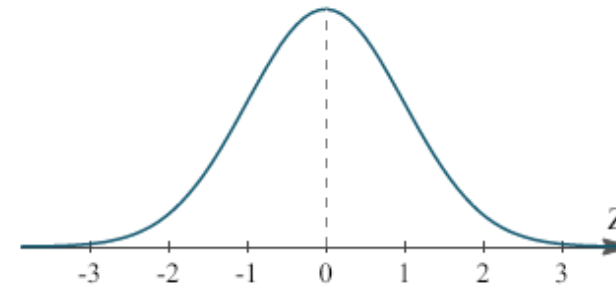
```
Out[4]: 0.040059156863817114
```

```
In [5]: 1-stats.norm.cdf(2.29)
```

```
Out[5]: 0.011010658324411393
```

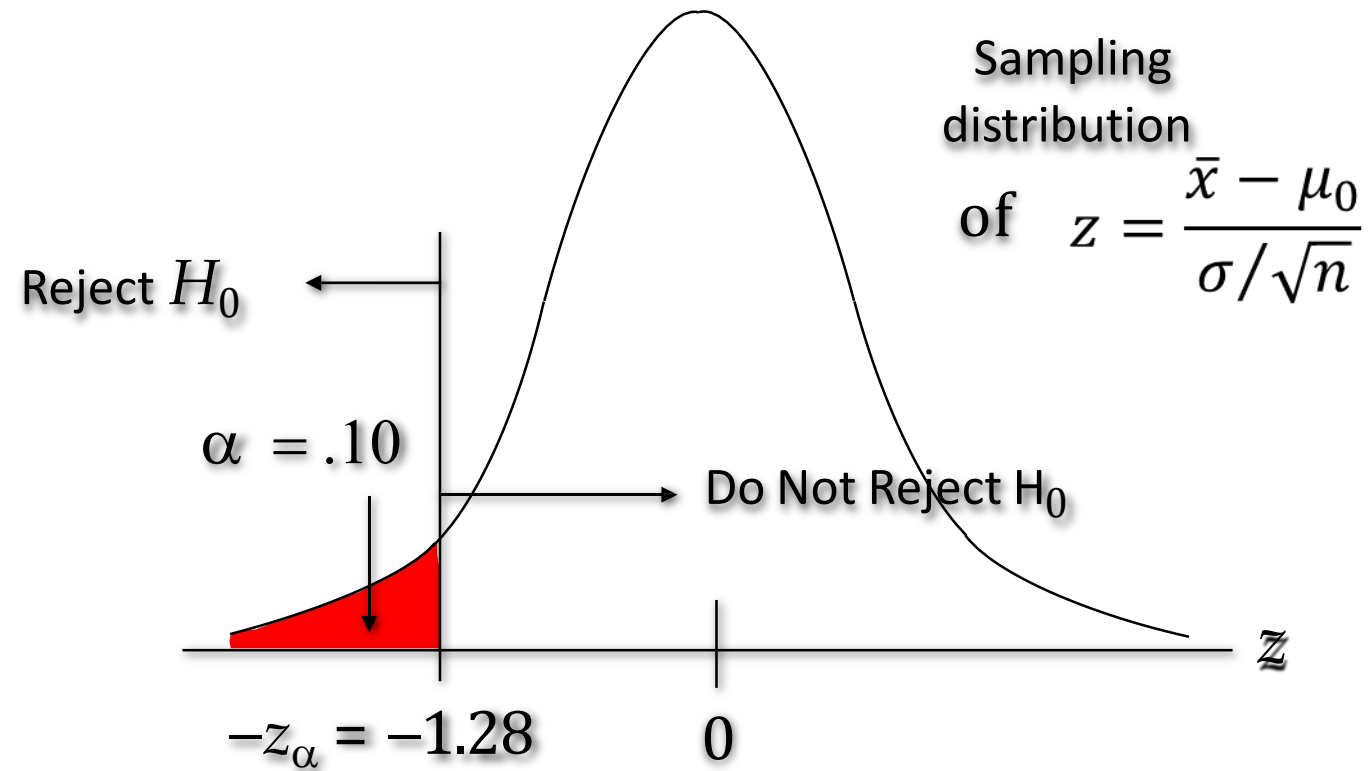
Critical Value Approach to One-Tailed Hypothesis Testing

- The test statistic z has a standard normal probability distribution.
- We can use the standard normal probability distribution table to find the z -value with an area of α in the lower (or upper) tail of the distribution.
- The value of the test statistic that established the boundary of the rejection region is called the critical value for the test.
- The rejection rule is:
Lower tail: Reject H_0 if $z \leq -z_\alpha$
Upper tail: Reject H_0 if $z \geq z_\alpha$



Lower-Tailed Test About a Population Mean: σ Known

Critical Value Approach

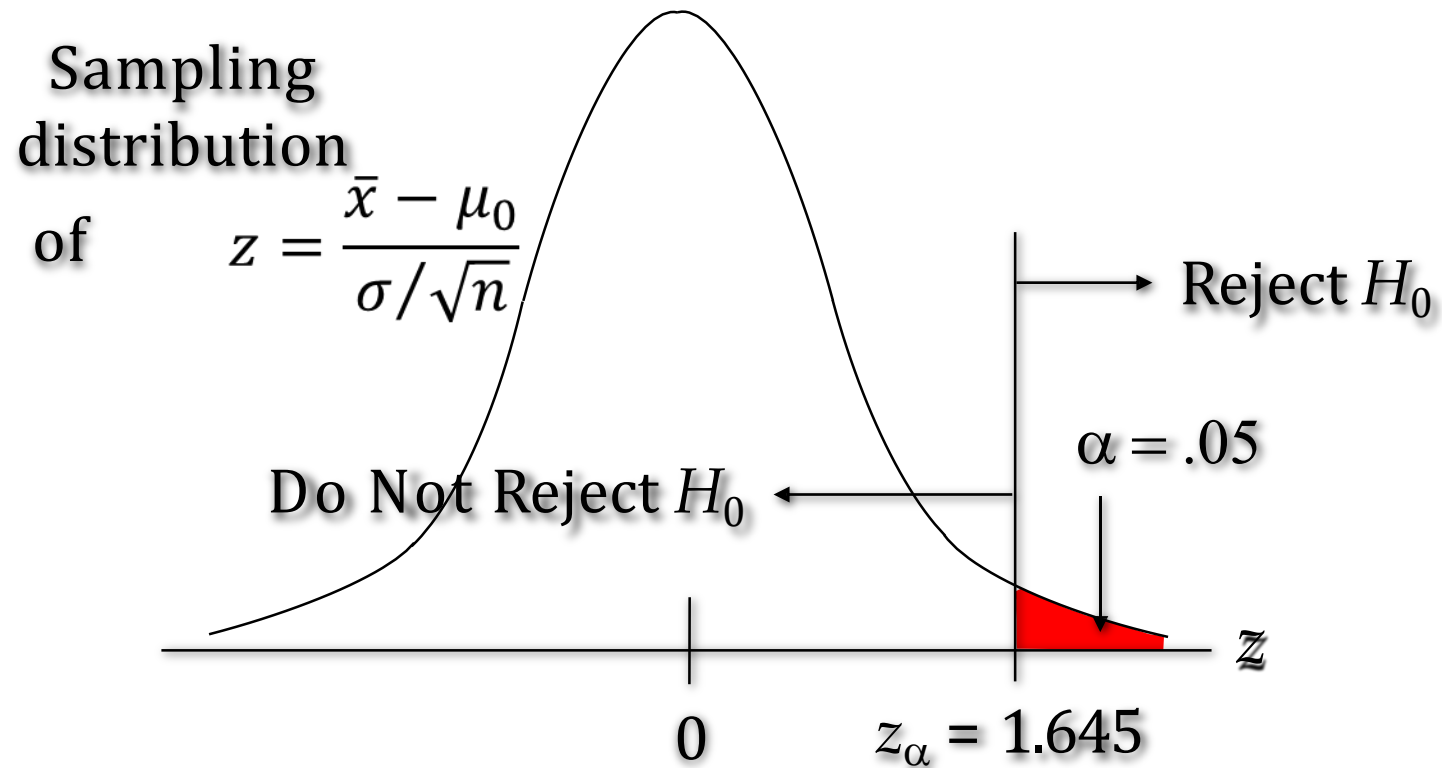


```
In [6]: stats.norm.ppf(0.1)
```

```
Out[6]: -1.2815515655446004
```

Upper-Tailed Test About a Population Mean: σ Known

Critical Value Approach



```
In [7]: stats.norm.ppf(0.95)
```

```
Out[7]: 1.6448536269514722
```

Steps of Hypothesis Testing – P value approach

- Step 1. Develop the null and alternative hypotheses.
- Step 2. Specify the level of significance α .
- Step 3. Collect the sample data and compute the test statistic.
- p-Value Approach
- Step 4. Use the value of the test statistic to compute the p -value.
- Step 5. Reject H_0 if $p\text{-value} \leq \alpha$.



Steps of Hypothesis Testing

Critical Value Approach

- Step 4. Use the level of significance α to determine the critical value and the rejection rule.
- Step 5. Use the value of the test statistic and the rejection rule to determine whether to reject H_0 .

One-Tailed Tests About a Population Mean:

σ

- Example: The mean response times for a random sample of 30 Pizza Deliveries is 32 minutes
- The population standard deviation is believed to be 10 minutes.
- The pizza delivery services director wants to perform a hypothesis test, with $\alpha = 0.05$ level of significance, to determine whether the service goal of 30 minutes or less is being achieved.



Given Values

- Sample
- Sample mean = 32 Min
- Sample size = 30
- Population
- $\alpha = 0.05$
- Population mean = 30 Min

p -Value Approach

One-Tailed Tests About a Population Mean: σ Known

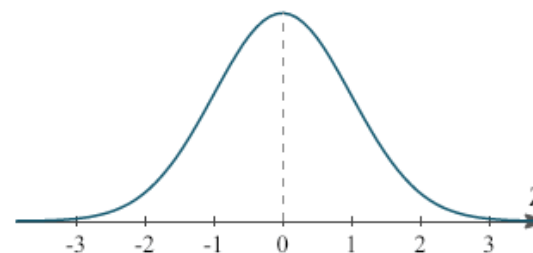
1. Develop the hypotheses.
2. Specify the level of significance.
3. Compute the value of the test statistic.

$$H_0: \mu \leq 30$$

$$H_a: \mu > 30$$

$$\alpha = .05$$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{32 - 30}{10 / \sqrt{30}} = 1.09$$



```
In [8]: 1-stats.norm.cdf(1.09)
```

```
Out[8]: 0.1378565720320355
```

One-Tailed Tests About a Population Mean:

σ

Known

p –Value Approach

4. Compute the p –value.

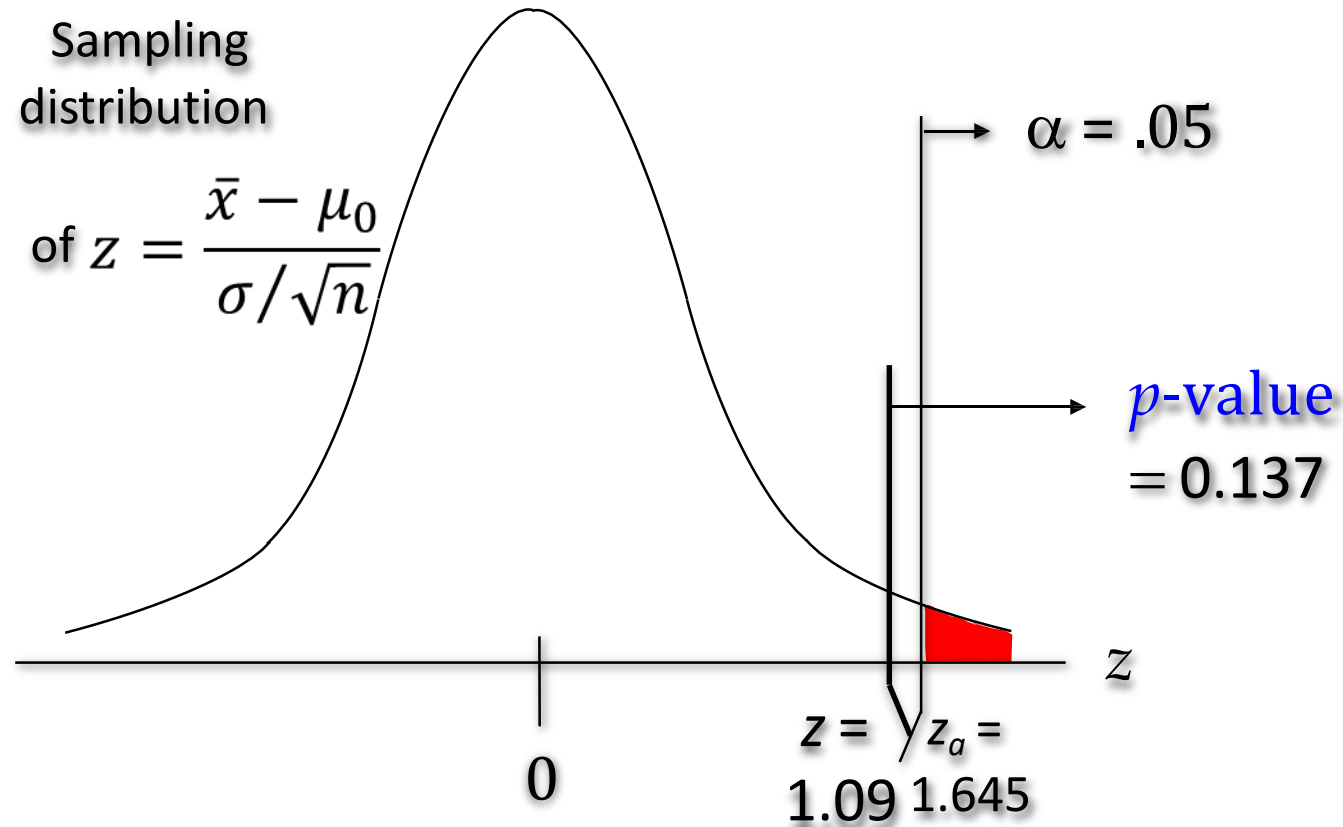
For $z = 1.09$, $p\text{-value} = 0.137$

5. Determine whether to reject H_0 .

- Because $p\text{-value} = 0.137 > \alpha = .05$, we do not reject H_0 .
- There are not sufficient statistical evidence to infer that Pizza delivery services is not meeting the response goal of 30 minutes.

One-Tailed Tests About a Population Mean: σ Known

p -Value Approach



Critical Value Approach

One-Tailed Tests About a Population Mean: σ Known

Critical Value Approach

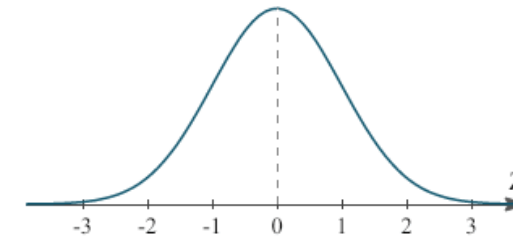
4. Determine the critical value and rejection rule.

– For $\alpha = .05$, $z_{.05} = 1.645$

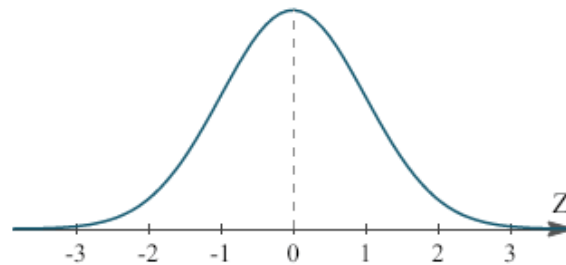
– Reject H_0 if $z \geq 1.645$

5. Determine whether to reject H_0 .

– Because $1.645 \geq 1.05$, we do not reject H_0 .



p-Value Approach to Two-Tailed Hypothesis Testing



Compute the p-value using the following three steps:

1. Compute the value of the test statistic z .
2. If z is in the upper tail ($z > 0$), find the area under the standard normal curve to the right of z .
3. If z is in the lower tail ($z < 0$), find the area under the standard normal curve to the left of z .
4. Double the tail area obtained in step 2 to obtain the p -value.

The rejection rule:

Reject H_0 if the p -value $\leq \alpha$.

Critical Value Approach to Two-Tailed Hypothesis Testing

- The critical values will occur in both the lower and upper tails of the standard normal curve.
- Use the standard normal probability distribution table to find $z_{\alpha/2}$ (the z -value with an area of $\alpha/2$ in the upper tail of the distribution).
- The rejection rule is:

Reject H_0 if $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$.

Two-Tailed Tests About a Population Mean: σ Known

- Example: Milk Carton
- Assume that a sample of 30 milk carton provides a sample mean of 505 ml.
- The population standard deviation is believed to be 10 ml.
- Perform a hypothesis test, at the 0.03 level of significance, population mean 500 ml and to help determine whether the filling process should continue operating or be stopped and corrected.



Given Values

- Sample
- Sample size = 30
- Sample mean = 505 ml
- Population
- Population mean = 500 ml
- Standard deviation = 10 ml
- Significance level 0.03

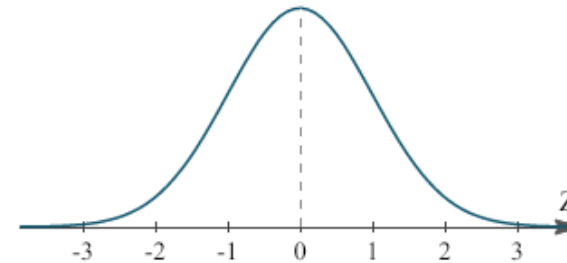
p –Value approach

Two-Tailed Tests About a Population Mean: σ Known

1. Determine the hypotheses.
2. Specify the level of significance.
3. Compute the value of the test statistic.

$$\begin{aligned}H_0: \mu &= 500 \\H_a: \mu &\neq 500 \\ \alpha &= .03\end{aligned}$$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{505 - 500}{10 / \sqrt{30}} = 2.74$$



```
In [9]: 1-stats.norm.cdf(2.74)
```

```
Out[9]: 0.003071959218650444
```

```
In [10]: (1-stats.norm.cdf(2.74))*2
```

```
Out[10]: 0.006143918437300888
```

Two-Tailed Tests About a Population Mean: σ Known

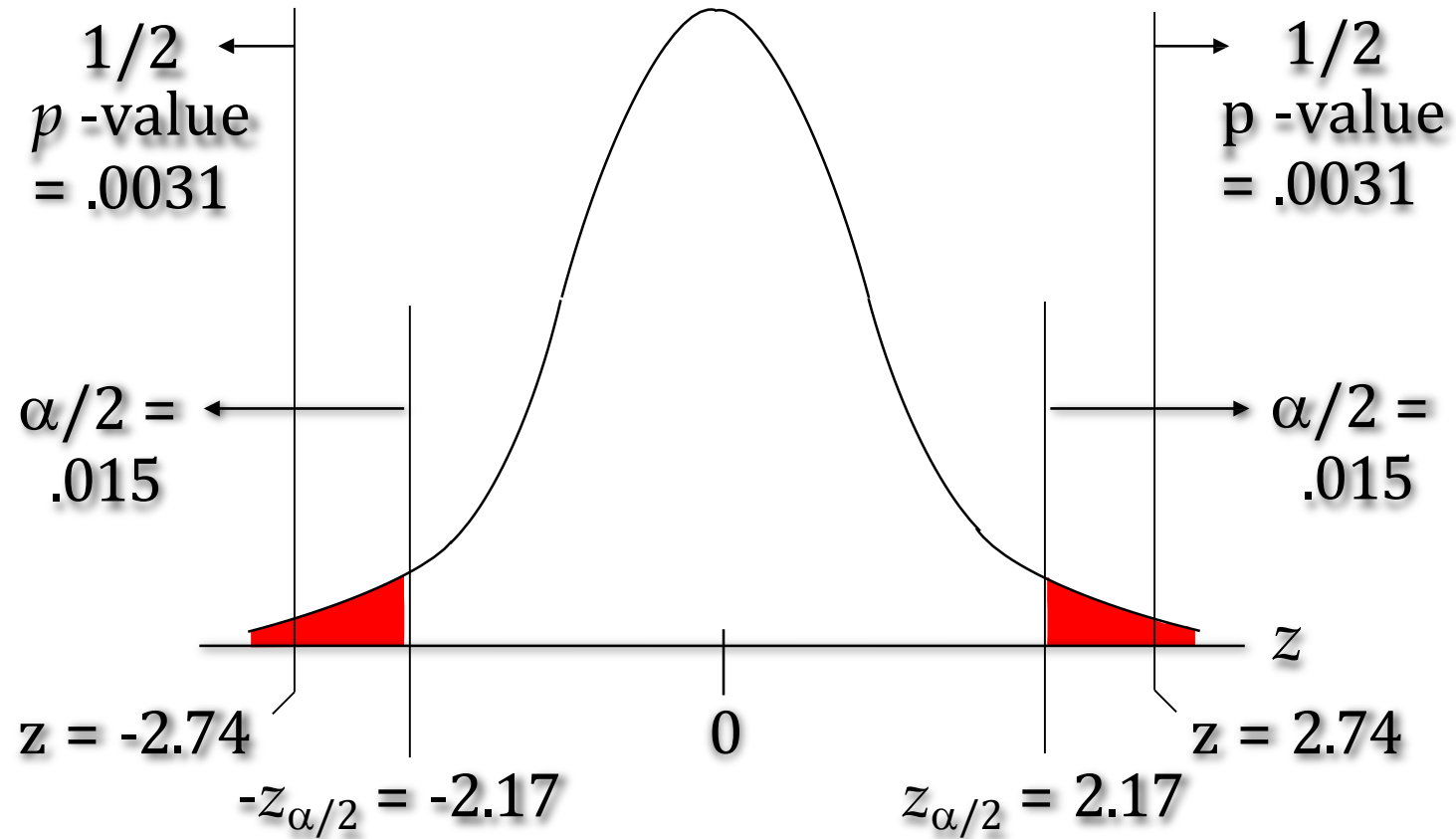
p -Value Approach

4. Compute the p -value.
 - For $z = 2.74$, $p\text{-value} = 2(1 - .9969) = .0061$
5. Determine whether to reject H_0 .
 - Because $p\text{-value} = .0062 < \alpha = .03$, we reject H_0 .

There is no sufficient statistical evidence to infer that the null hypothesis is true (i.e. the mean filling quantity is not 500 ml)

Two-Tailed Tests About a Population Mean: σ Known

p-Value Approach



Critical Value Approach

Two-Tailed Tests About a Population Mean

σ

- Critical Value Approach

4. Determine the critical value and rejection rule, for $\alpha/2 = .03/2 = .015$, $z_{.015} = 2.17$

Reject H_0 if $z < -2.17$ or $z > 2.17$

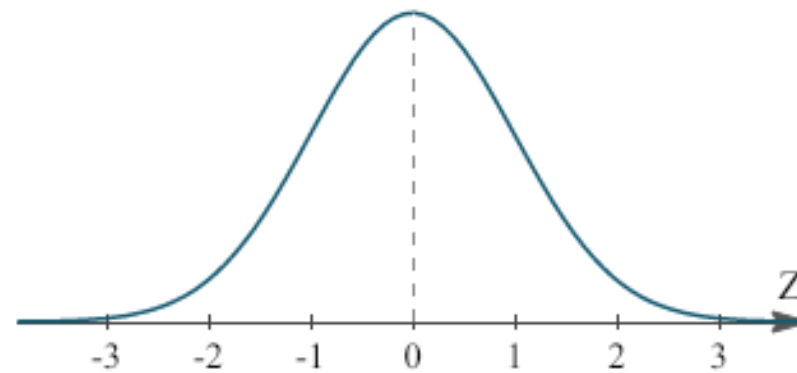
5. Determine whether to reject H_0 .

Because $2.74 > 2.17$, we reject H_0 .

There is sufficient statistical evidence to infer that the null hypothesis is not true

In [12]: `stats.norm.ppf(0.015)`

Out[12]: -2.1700903775845606

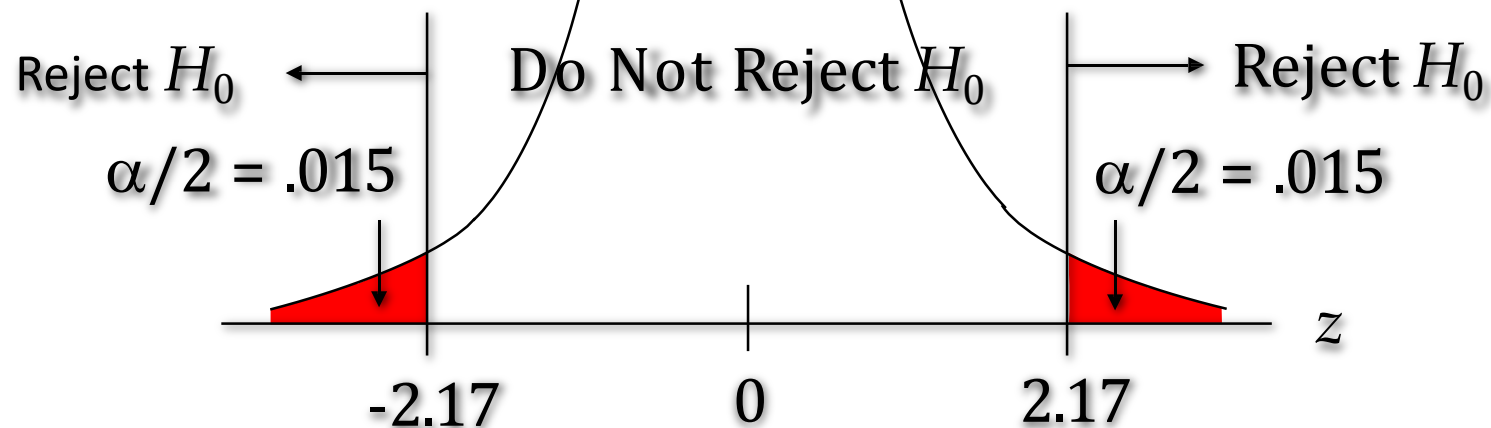


Two-Tailed Tests About a Population Mean : σ Known

Critical Value Approach

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{505 - 500}{10 / \sqrt{30}} = 2.74$$

Sampling
distribution
of $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$



T test



T test

Parameter – Degrees of Freedom

The t distribution has only one parameter, the degrees of freedom (DF). In t-tests, DF are linked to the sample size. For 1-sample and paired t-tests, $DF = N - 1$. For 2-sample t-tests, it equals $N - 2$. Hence, as the sample size increases, the DF also increases.

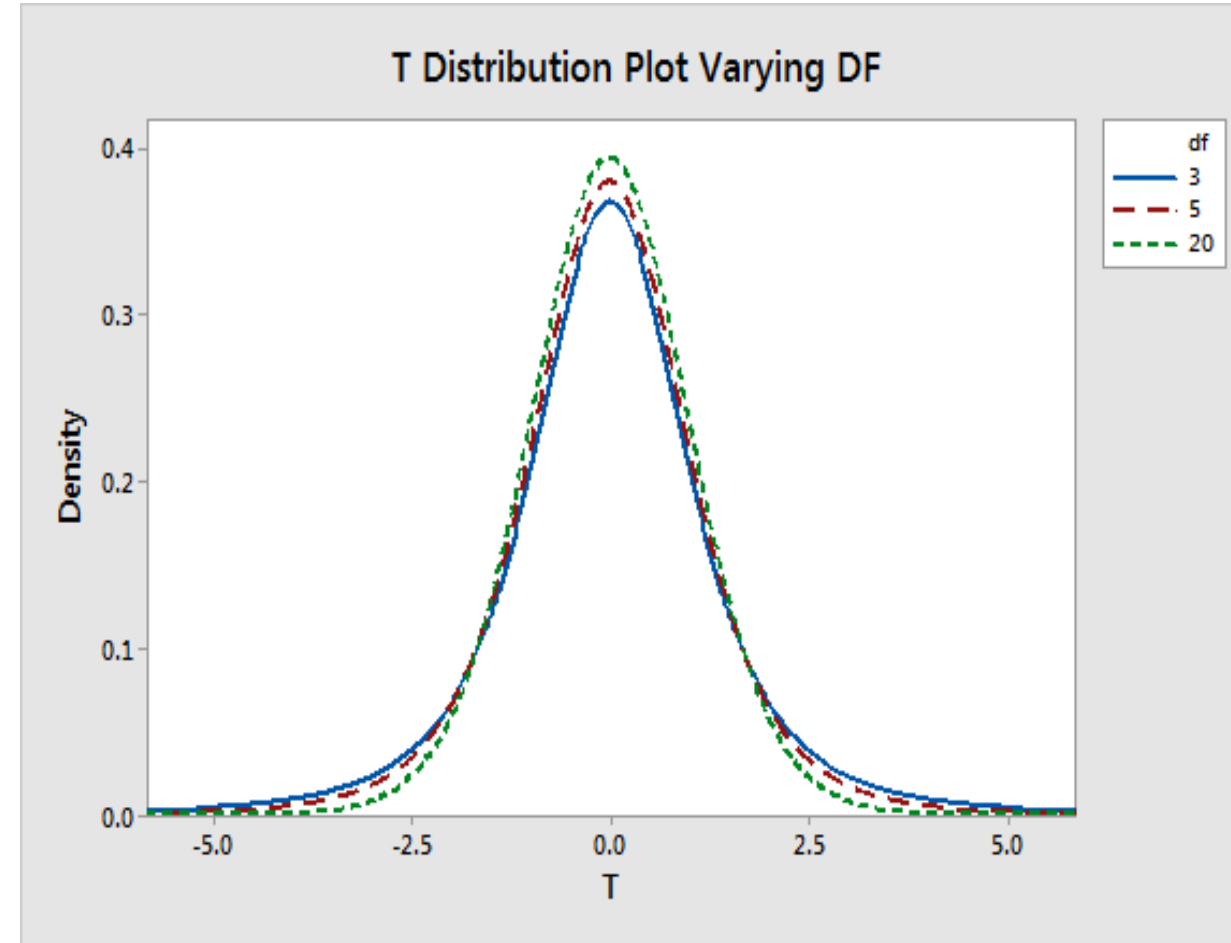
What are Degrees of Freedom?

The degrees of freedom (DF) in statistics indicate the number of independent values that can vary in an analysis without breaking any constraints.



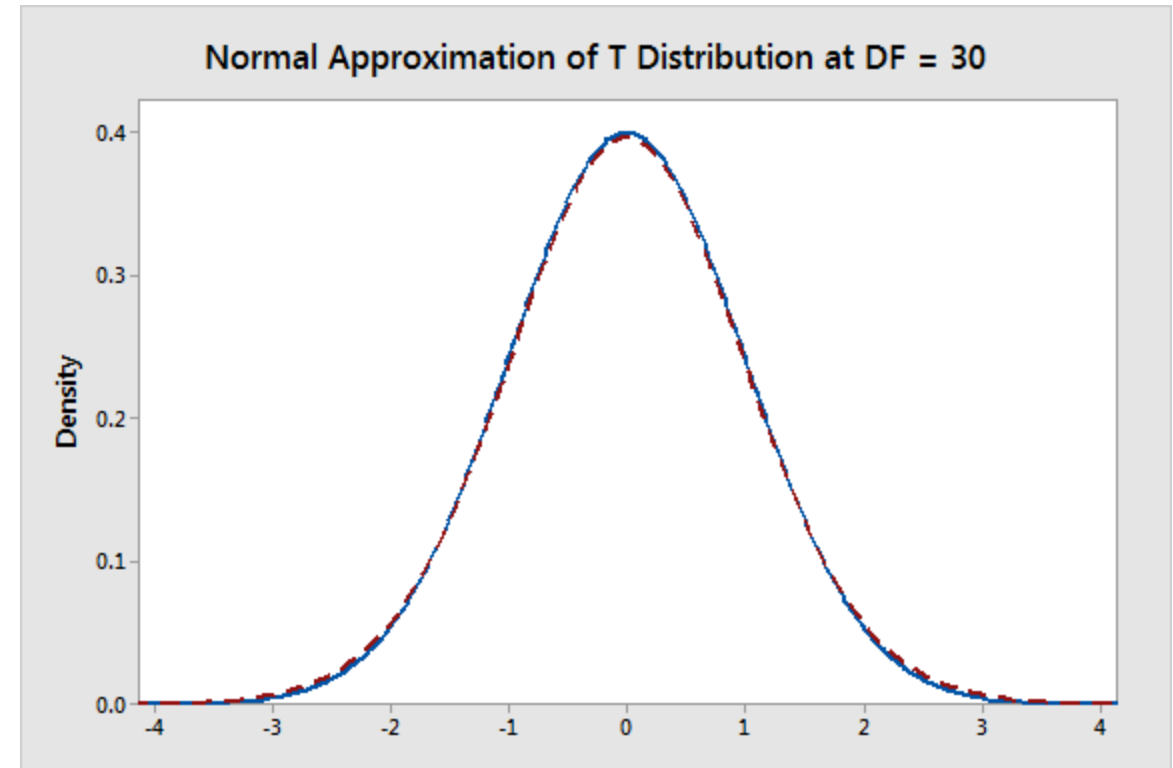
Degree of freedom in t test

- T tests are hypothesis tests for the mean and use the t-distribution to determine statistical significance.
- A 1-sample t test determines whether the difference between the sample mean and the null hypothesis value is statistically significant. For a 1-sample t test, use $n - 1$ to calculate degrees of freedom.
- As the degrees of freedom increase, the curve pulls in tighter around zero—the tails become thinner and the peak becomes taller.



Degree of freedom in t test

- At around 30 degrees of freedom, the t distribution closely approximates the standard normal distribution (Z-distribution), as shown below. Consequently, when your sample size exceeds ~ 30 , t-tests and Z-tests provide very similar results.



Tests About a Population Mean: σ Unknown

Rejection Rule: p -Value Approach

Reject H_0 if p -value $\leq \alpha$

Rejection Rule: Critical Value Approach

$H_0: \mu \geq \mu_0$ Reject H_0 if $t \leq -t_\alpha$

$H_0: \mu \leq \mu_0$ Reject H_0 if $t \geq t_\alpha$

$H_0: \mu = \mu_0$ Reject H_0 if $t \leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$

```
In [10]: from scipy import stats  
import numpy as np
```

```
In [11]: x=[10,12,20,21,22,24,18,15]  
stats.ttest_1samp(x,15)
```

```
Out[11]: Ttest_1sampResult(statistic=1.5623450931857947, pvalue=0.1621787560592894)
```

One-Tailed Test About a Population Mean:

- In an ice cream parlor at IIT Roorkee, the following data represent the number of ice-creams sold in 20 days

Unknown Demand Example

- Test hypothesis $H_0: \mu \leq 10$
- Use $\alpha = .05$ to test the hypothesis.



Day	No. of Ice-cream Sold	Day	No. of Ice-cream Sold
1	13	11	12
2	8	12	11
3	10	13	11
4	10	14	12
5	8	15	10
6	9	16	12
7	10	17	7
8	11	18	10
9	6	19	11
10	8	20	8

```
In [8]: x=[13,8,10,10,8,9,10,11,6,8,12,11,11,12,10,12,7,10,11,8]
```

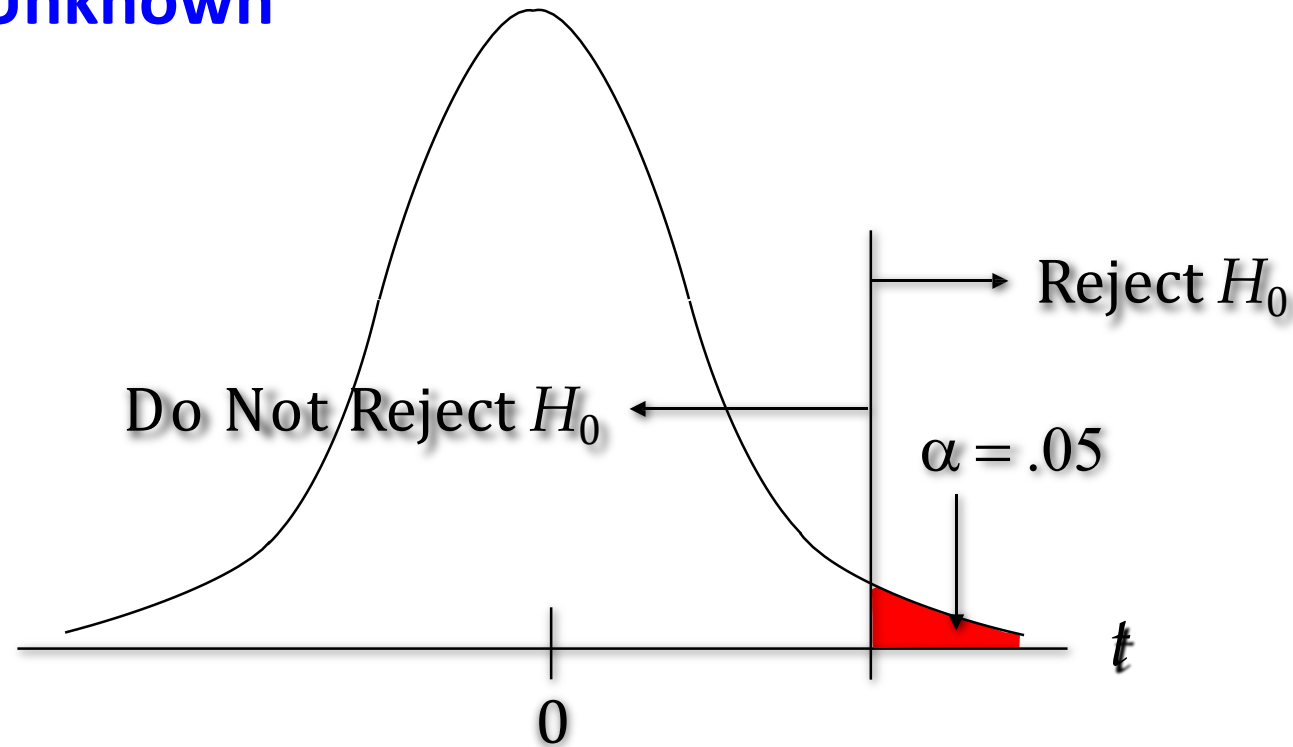
```
In [9]: stats.ttest_1samp(x,10)
```

```
Out[9]: Ttest_1sampResult(statistic=-0.35843385854878496, pvalue=0.7239703579964252)
```

```
In [10]: 0.7239703579964252/2
```

```
Out[10]: 0.3619851789982126
```

One-Tailed Test About a Population Mean: σ Unknown



In [3]: `stats.t.cdf(-0.384,19)`

Out[3]: 0.35262102566795583

[2]: `stats.t.ppf(0.05,19)`

Out[2]: -1.7291328115213678